MILITARY SPECIFICATION

AMPLIFIER, CONTROL, SINGLE CHANNEL, S-2 COMPASS

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope.- This specification covers certain design requirements and all performance requirements for procurement of one type single channel electronic control amplifier.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-M-151	Metals, General Specification for inspection of
QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-C-5015	Connector, Electrical, "AN" Type
MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-C-6054	Container, Steel Shipping
MIL-D-5028	Drawings and Data Lists: Preparation of Manufacturers' (For Production Aircraft, Guided Missiles, Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5400(ASG)	Electronic Equipment, Aircraft, General Specification for
MIL-I-6181	Interference Limits and Tests, Aircraft Electrical and Electronic Equipment
MIL-P-6064	Packaging of Lightweight Aircraft Accessories
MIL-P-6906	Plates, Information and Identification
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-C-19938(Aer)	Control Directional Electrically Driven Gyro S-2 Compass and S-5 Automatic Pilot

STANDARDS

Military

MS28082(Aer)	Amplifier, Electronic Control, Single Channel, S-2 Compass, Type
MIL-STD-130	Identification Markings of U. S. Military Property

DRAWINGS

Air Force - Navy Aeronautical Standard Drawings

AND 10398	Metals, Definition of Dissimilar

PUBLICATIONS

Air Force - Navy Aeronautical Specification Bulletin

No. 143	Specification and Standards, Use of
No. 405	Storage Life - Aeronautical Articles

FSC - 6605

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(When requesting any of the applicable documents refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 700 Robbins Avenue, Philadelphia 11, Pennsylvania, Attention: Code ODPT).

3. REQUIREMENTS

3.1 Qualification.- The amplifier furnished under this specification shall be a product which has been tested and has passed the Qualification tests specified herein.

3.2 Materials.- Materials shall conform to applicable specifications and as specified herein. Materials which are not covered by applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Critical materials.- Noncritical materials shall be used where practicable. Where the use of a critical material is essential to meet specification requirements, the material used shall be the least critical of those which are adequate for the purpose.

3.2.2 Metals.- Metal shall be of the corrosion-resistant type, or suitably protected to resist corrosion in salt spray or atmospheric conditions to which the amplifier may be subjected when in storage or during normal service life.

3.2.2.1 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in Drawing AND10398.

3.2.3 Nonmagnetic materials.- Nonmagnetic materials shall be used for all parts of the amplifier, except where magnetic materials are essential.

3.2.4 Fungus-proof materials.- Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials which are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Bureau of Aeronautics.

3.2.5 Selection of materials.- Specifications and Standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.5.1 Standard parts.- AN or MS parts shall be used wherever they are suitable for the purpose, and shall be identified by their numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they have suitable properties and are replaceable by the AN or MS parts without alteration, and provided the corresponding AN or MS part numbers are referenced in the parts lists, and, if practicable, on the contractor's drawings. In applications for which no suitable corresponding AN or MS part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.3 Design and construction.- The outline dimensions of the amplifier shall conform to the requirements of MS28082(Aer). The amplifier shall be designed to amplify the error signal of the directional gyro control conforming to Specification MIL-C-19938(Aer) and to detect its phase to control the direction and amount of torque in the slaving torque motor. The design shall be reasonably simple to permit overhaul or repair without requiring special tools or fixtures. The amplifier shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.3.1 Slaving channel.- The amplifier shall contain a slaving channel to receive a signal of controlled low amplitude phase reversing voltage and shall be capable of producing a phase reversing alternating current voltage of required power as indicated under Slaving output and polarity tests specified herein.

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3.3.2 Electronic parts.- Electronic parts and the application thereof shall be in accordance with Specification MIL-E-5400. Parts that do not appear on approved lists in accordance with Specification MIL-E-5400 shall not be used unless approved by the Bureau of Aeronautics.

3.3.3 Electrical receptacle.- The electrical receptacle used shall conform to Specification MIL-C-5015 and shall be of the type specified on MS . The receptacle shall be wired as shown in Figure 1.

3.3.4 Electrical power.- The amplifier shall be electrically operated from 115 volt, 3 phase, 400 cycle alternating current and 28 volts direct current power.

3.3.5 Protective device.- A 1/4 to 1/2 ampere dual element fuse shall be provided in the lead to the primary of the power and transmitter excitation transformer to protect against overload conditions.

3.3.6 Case.- The case shall be made of non-ferrous low density metal and shall be entirely covered with a durable dull black finish.

3.4 Performance.- The amplifier shall meet the performance requirements specified in Section 4, when subjected to the applicable tests.

3.5 Interchangeability.- All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Special tools.- The design of the amplifier shall be such as to accommodate to the greatest possible extent, disassembly, re-assembly, and service maintenance by those tools and items of maintenance equipment which are normally available as commercial standards.

3.7 Physical requirements.-

3.7.1 Weight.- The weight of the amplifier shall not exceed 6-3/4 pounds.

3.7.2 Finish.- Protective coatings and finishes which will crack, chip or scale during normal service life or due to extremes of atmospheric conditions shall not be used.

3.7.2.1 Aluminum alloy parts.- Where practicable, aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.7.2.2 Iron and steel parts.- Where practicable, iron and steel parts shall be cadmium plated in accordance with Specification QQ-P-416.

3.7.3 Screw threads.- Screw threads 0.060 inch or larger in diameter shall be in accordance with Specification MIL-S-7742.

3.7.4 Soldering.- Soldering shall be performed in accordance with Specification MIL-S-6872.

3.8 Marking.- All markings shall be durable to withstand usage encountered in service.

3.8.1 Electrical receptacle code number.- The electrical receptacle code number markings as specified on the applicable military standard shall be legibly and permanently applied to the amplifier case or electrical receptacle.

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3.8.2 Nameplate.- The nameplate shall conform to Specification MIL-P-6906, shall be securely attached to the instrument case, and shall be legibly and permanently marked with the following information in accordance with MIL-STD-130.

BUAER - U. S. Navy
Amplifier, Electronic Control, Single Channel, S-2 Compass
MS Part No.
Stock No. (Govt.)
Manufacturer's Part No.
Manufacturer's Serial No.
Contract No. (Or Order No. if Contract No. is not assigned)
Manufacturer's Name or Trademark

3.8.2.1 Manufacturer's part no.- The manufacturer's part number marked in the space provided in the nameplate shall be identical with the manufacturer's engineering production drawing number including applicable dash numbers if the drawing is tabulated and covers more than one part.

3.9 Installation.-

3.9.1 Installation instructions.- The contractor shall furnish with each amplifier one printed copy of instructions, with illustrations and diagrams, if necessary, covering the installation of the amplifier. Prior to printing, two copies shall be furnished to the Bureau of Aeronautics for approval. The instructions shall be printed on 8-1/2 by 11 or 11 by 17 inch paper.

3.9.2 Envelope.- An envelope furnished by the contractor containing installation instructions shall be packaged with each amplifier, and shall be marked with the following information:

"IMPORTANT"

"THIS ENVELOPE CONTAINS INSTRUCTIONS"

3.10 Workmanship.- All details of workmanship shall be in accordance with high grade aircraft instrument manufacturing practice. The amplifier, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding and brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of tests.- The inspection and testing of the amplifier shall be classified as follows:

a. Qualification tests: Qualification tests are those tests performed on amplifier submitted for qualification as a satisfactory product.

b. Acceptance tests: Acceptance tests are those tests performed on amplifier manufactured and submitted for acceptance under contract.

4.2 Qualification tests.- The qualification tests of the amplifier shall consist of all tests of this specification and inspection. The tests shall be conducted in the following sequence:

Individual tests
Sampling tests
Radio noise interference
High altitude-Low temperature operation
Low temperature exposure
High temperature exposure
Humidity
Fungus resistance
Salt spray

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4.2.1 Qualification test samples.- Qualification test samples shall consist of two amplifiers of each manufacturer's part number submitted as samples for testing. After obtaining authorization for submittal from the Bureau of Aeronautics, the qualification test samples shall be forwarded to the Supply Officer, Naval Air Development Center, Johnsville, Pennsylvania, Attention: Director, Aeronautical Instruments Laboratory. Each sample shall be plainly identified by a securely attached durable tag marked with the following information:

Sample for qualification test
Submitted by (Name) (Date) for qualification tests in accordance with the requirements of Specification MIL-A- under authorization (reference letter authorizing test)
Amplifier, Electronic Control, Single Channel, S-2 Compass, Type
Manufacturer's Part No.
Name of Manufacturer

4.3 Acceptance tests.- The acceptance tests of the amplifier shall consist of the individual tests and the sampling tests of this specification.

4.3.1 Individual tests.- The individual tests of the amplifier shall consist of the following tests conducted on each amplifier:

Inspection
Output voltages
Slaving output and polarity
Grounded signal

4.3.2 Sampling tests.- The sampling tests of the amplifier shall consist of the following tests conducted on each sampling test sample:

Low temperature operation
High temperature operation

4.3.2.1 Sampling test samples.- Sampling test samples shall consist of at least one amplifier selected at random by the inspector from each lot of 100 or less amplifiers which have passed the individual tests. A lot shall consist of identical amplifiers with the same manufacturer's part number, manufactured under substantially the same conditions and submitted at substantially the same time. The inspector shall select as many amplifiers as considered necessary to ensure compliance of the entire lot with the requirements of these tests. Amplifiers passing the sampling tests shall not be delivered on contract until they have been rebuilt and resubmitted to all individual tests.

4.4 Test conditions.-

4.4.1 Atmospheric conditions.- Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and at a temperature of approximately 25°C (77°F). When tests are made with either, or both, the atmospheric pressure and temperature substantially different from the above values, proper allowance shall be made for the difference from specified conditions.

4.4.2 Power supply.- Unless otherwise specified the amplifier shall be energized during all tests with 27.5 ± 0.5 volts direct current and 115 ± 2 volts, 400 ± 5 cycles, 3 phase alternating current.

4.5 Test methods.-

4.5.1 Inspection.- Each amplifier shall be inspected to determine compliance with the requirements specified herein with respect to Materials, Workmanship, and Marking.

4.5.2 Output voltages.- The designated output voltages shall be obtained from the amplifier between the points stipulated in Table 1, under the following conditions:

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TABLE I

	FAST SLAVING	NORMAL SLAVING
Transmitter Excitation Phases A and C	23V ac ±5%	23V ac ±5%
Fixed Field Excitation Phases A and C	115V ac ±5%	115V ac ±5%
Leveling Control Field Phases B and A Through internal isolation transformer, one lead of output grounded	30V ac ±5%	15V ac ±5%

The referenced fast slaving voltages shall occur immediately upon applying power to the amplifier. The normal slaving voltages shall occur between 2 and 3-1/2 minutes after power is applied to the amplifier. This test shall be preceded by at least a 15 mimute period of inoperation to allow the time delay relay to normalize.

4.5.3 Slaving output and polarity.-

4.5.3.1 Fast slaving output.- The slaving control field output voltage, when measured across a 3500 ohm load for the condition of fast slaving with an input of 0.4V of both in phase and 180 degrees out of phase signal, shall be between 65 and 85V. In addition, the difference between output voltages for in phase and 180 degrees out of phase input signal voltage of 0.4V shall not exceed 6V.

4.5.3.2 Normal slaving output.- The slaving control field output voltage limits, when measured across a 3500 ohm load for the condition of normal slaving, shall be as indicated below when the input signal voltage is as stipulated. In addition, the unbalance between output voltages for in phase and 180° out of phase input signal voltages of the same magnitude shall be as specified in Table II.

TABLE II

Input Signal Volts	Output Signal Volts	Maximum Unbalanced Volts
0.04	2.6 to 7.0	0.5
0.08	5.2 to 9.0	1.3
0.20	6.8 to 9.2	1.3
0.40	6.6 to 9.4	1.3

4.5.3.3 Polarity.- With an input signal voltage from the high side with respect to ground in phase with the 800 cycles reference voltage high side with respect to ground, the output voltage, as measured across a 3500 ohm load, from the high side with respect to ground, shall be in phase with the line voltage "B" phase with respect to "A" phase.

4.5.4 Grounded signal.- The amplifier shall be operated under normal slaving conditions, except that the signal input terminal which is normally above ground shall be connected to the ground. The output as measured across a 3500 ohm load shall not exceed 2V.

4.5.5 Low temperature operation.- The amplifier shall be subjected to a temperature of -55°±2°C (-67° ± 3.6°F) for a period of 4 hours. At the end of this period and while still at this temperature, the amplifier shall be subjected to all the individual tests and the same tolerances shall apply, except that the grounded signal tolerance shall not exceed 3 volts.

4.5.6 High temperature operation.- The amplifier shall be subjected to a temperature of 70 ± 2°C (158° ± 3.6°F) for a period of 4 hours. At the end of this period and while still at this temperature, the amplifier shall be subjected to all of the individual tests. The same tolerance shall apply except that the grounded signal tolerance shall not exceed 3 volts.

4.5.7 Radio interference.- The amplifier shall be operated on test voltage specified herein. The amplifier shall then be tested for conducted and radiated radio noise interference in accordance with Specification MIL-I-6181.

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4.5.8 High altitude-Low temperature operation.- The amplifier shall be energized from the standard power supply and shall be subjected to a pressure equivalent to 50,000 ± 500 feet at a temperature of -55° ± 2°C (-67° ± 3.6°F) for a period of 4 hours. While at this temperature and pressure the amplifier shall be subjected to and shall meet the requirements of the Individual Tests.

4.5.9 Low temperature exposure.- The amplifier shall be subjected to a temperature of -65° ± 2°C (-85° ± 3.6°F) for a period of 48 hours. The temperature shall then be raised to -55° ± 2°C (-67° ± 3.6°F) for a period of 24 hours. After the chamber has been returned to room temperature and the amplifier has saturated at this temperature for a minimum period of 4 hours, it shall be subjected to and meet the requirements of the Individual Tests. There shall be no damage which would adversely affect subsequent operation.

4.5.10 High temperature exposure.- The amplifier shall be subjected to a temperature of 70° ± 2°C (158° ± 3.6°F) for a period of 24 hours. After the chamber has returned to room temperature and the amplifier has saturated at this temperature for a minimum period of 4 hours it shall be subjected to and shall meet the requirements of the Individual Tests. There shall be no damage which would adversely affect subsequent operation. This test may be combined with the High temperature operation test at the discretion of the testing agency.

4.5.11 Humidity.- The amplifier shall be mounted in a chamber and maintained at a temperature of 70° ± 2°C (158° ± 3.6°F) and a relative humidity of 95 ± 5 percent for a period of 6 hours. Distilled water shall be used to obtain the required humidity. At the end of the 6-hour period, the heat shall be shut off and the amplifier shall be allowed to cool for a period of 18 hours in this atmosphere, in which the humidity rises to 100 percent as the temperature decreases. This cycle shall be repeated a sufficient number of times to extend the total time of test to 120 hours (5 cycles). Within one hour after the end of the 120 hour (5 cycles) period, the amplifier shall be subjected to and meet the requirements specified in the Individual Tests. Examination for corrosion and other deterioration shall be made between 24 and 48 hours after completion of the humidity exposure. There shall be no corrosion or other deterioration which would adversely affect subsequent operation of the amplifier.

4.5.12 Fungus resistance.- The amplifier shall be placed in a mold chamber and shall be sprayed with a suspension of freely sporulating mixed spores similar to those encountered in tropical climates. Distilled water, with or without a wetting agent, shall be introduced directly into a stock culture of one fungus and shaken gently without disturbing the agar. The spore suspension shall then be sprayed through an atomizer over the test sample, or the test sample shall be dipped into the spore solution. Five groups of fungi are listed below and one fungus from each group shall be used:

Group I	Chaetomium globosum USDA 1042.4 or Myrothecium verrucaria USDA 1334.2
Group II	Rhizopus nigricans S.H. 32 or Aspergillus niger USDA -Tc215-4247
Group III	Aspergillus flavus AMC No. 26 or Aspergillus terreus PQMD 82J
Group IV	Penicillium luteum USDA 1336.1, Penicillium Sp. USDA 1336.2 or Penicillium citrinum ATCC 9848
Group V	Memoniella echinata - AMC No. 37 or Fusarium moniliforme USDA 1004.1

Substitutions for the above fungi may be made provided they are acceptable to the agency performing the qualification tests. The temperature of the chamber shall be maintained at 30 ± 2°C (86 ± 3.6°F) with a relative humidity of 95 ± 5 percent. The duration of this test shall not be less than 14 days. At the conclusion of this test, the amplifier shall be examined and there shall be no fungus growth which would adversely affect subsequent operation. At the option of the agency performing the qualification tests, tests on component parts of the instrument may be accepted in lieu of, or in addition to, tests on the assembled instrument.

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4.5.13 Salt spray.- The amplifier, with no power applied, but with external connections made to simulate installed conditions, shall be subjected to a salt spray in accordance with Specification QQ-M-151 for a period of 50 hours. At the end of the 50 hour period, the amplifier shall be examined and there shall be no corrosion or deterioration which would adversely affect subsequent operation of the amplifier. The amplifier shall be subjected to and shall meet the requirements specified in the Individual Tests.

4.6 Rejection and retests.- When any amplifier fails to meet the requirements of the sampling test, the lot represented shall be rejected and returned at the contractor's expense. Any amplifier failing to meet the requirements of the Individual Tests shall be rejected and returned at the contractor's expense. Amplifiers which have been rejected may be replaced or repaired to correct the defects and resubmitted for test. When this has been accomplished, all specified tests shall be repeated. Before resubmitting, full particulars concerning previous rejection and action taken to correct the original defects shall be furnished the inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring activity.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of Section 5 apply only to direct purchases by or direct shipment to the Government.

5.2 Preservation and packaging.- The preservation and packaging shall be in accordance with Specification MIL-P-6064, Group VII, Type I except that the size of the container shall conform to MIL-C-6054.

5.3 Packing.- Packing shall conform to Specification MIL-P-6064 for both domestic and overseas shipment.

5.4 Marking.- In addition to the markings required by Specification MIL-P-6064, each container shall be durably and legibly marked with the following information in such a manner that the markings will not become damaged when the containers are opened:

Amplifier, Electronic, Control, Single Channel, S-2 Compass
MS Part No. (Insert proper dash No.)
Date of Manufacture

5.5 Reinspection dates.- The reinspection date markings shall be in accordance with ANA Bulletin No. 405.

6. NOTES

6.1 Intended use.- The amplifier covered by this specification is intended for use in conjunction with the S-2 compass.

6.2 Ordering data.- Requisitions, contracts and orders shall specify the type and part number of the amplifier desired, the quantity, and whether overseas packing is to be furnished. Procuring documents should specify the level of packaging and packing desired.

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the bid opening date, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief, Bureau of Aeronautics, Department of the Navy, Washington 25, D.C.

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